








# FOOD EXPIRY DETECTION AND REMINDER SYSTEM

## AI-POWERED FOOD INVENTORY MANAGEMENT

## MINI PROJECT REPORT

Submitted to the

**SCHOOL OF ENGINEERING**

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**BACHELOR OF TECHNOLOGY**

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# BONAFIDE CERTIFICATE

Certified that this project report entitled "FOOD EXPIRY DETECTION AND REMINDER SYSTEM" is the bonafide work of Karunya S, Abarna V and Yogalakshmi M, who carried out the project under the guidance and supervision of Ms. S. M. Sowmiya, Department of Artificial Intelligence and Machine Learning, Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai.

Certified further that, to the best of our knowledge, the work reported here has been prepared for academic purposes and does not form part of any other project report or dissertation submitted for a degree or award.

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|---|---|
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# DECLARATION

We hereby declare that the mini project report entitled " **FOOD EXPIRY DETECTION AND REMINDER SYSTEM**", submitted to Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence), is a record of original work carried out by us under the guidance of **Ms. P. Mohana Priya, M.E, (Ph.D.),** Assistant Professor, Department of Artificial Intelligence and Machine Learning. The work has not been submitted, in part or in full, for the award of any other degree or diploma.

Place: Chennai

Date: 30.10.2026

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# ABSTRACT

Food waste is a growing concern, partly because people forget the expiry dates of food stored at home or in small businesses. Packaged products generally carry printed date information, while fresh items such as fruits and vegetables may not have a label. This project proposes a Food Expiry Detection and Reminder System that combines barcode scanning, Optical Character Recognition (OCR), shelf-life estimation and inventory notifications in one application.For packaged food, a barcode can help identify the product and OCR can extract manufacturing or expiry dates from the label. If the date is missing or cannot be read, the system uses a predefined food-type dataset and storage-condition inputs to estimate an indicative shelf-life window. Users can customize expiry durations for individual items, such as assigning a separate value to tomatoes instead of relying only on a broad vegetable category. Inventory records store the item name, expiry date, remaining shelf life and status, while reminders notify users before the expected expiry date.The proposed implementation is organized as a web-based application using React, TypeScript and Tailwind CSS for the frontend; Python and FastAPI for backend services; MongoDB Atlas for data storage; and OCR and image-processing libraries for scanning. The report adapts the research and technical direction of the supplied food-expiry reference papers to a mini-project report structure. The cited base study reported 72.9% product-identification success and 83.3% OCR date-reading success on its evaluation sample; these are reference-paper results, not measured results of the proposed project. The system is intended to support inventory awareness and reduce avoidable waste. Shelf-life estimates are indicative and must not replace package instructions or food-safety guidance.

Keywords: Food Expiry Detection, OCR, Barcode Scanning, Shelf-Life Estimation, Food Inventory, Reminder System, Food Waste Reduction.

# LIST OF FIGURES


| FIGURE NO. | TITLE |
|---|---|
| Figure 1 | Proposed system architecture |
| Figure 2 | Food expiry detection workflow |
| Figure 3 | Shelf-life estimation decision flow |
| Figure 4 | Inventory and reminder workflow |

# LIST OF ABBREVIATIONS

| ABBREVIATION | FULL FORM |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| CNN | Convolutional Neural Network |
| CRUD | Create, Read, Update and Delete |
| CV | Computer Vision |
| DB | Database |
| HTTP | Hypertext Transfer Protocol |
| ML | Machine Learning |
| OCR | Optical Character Recognition |
| UI | User Interface |
| UX | User Experience |
| URL | Uniform Resource Locator |
| JWT | JSON Web Token |
| REST | Representational State Transfer |
| OCR | Optical Character Recognition |
| ETA | Estimated Time of Arrival / Estimated Time to Expiry (context-dependent) |

# CHAPTER 1. INTRODUCTION

## 1.1 OVERVIEW OF THE PROJECT

### Introduction

Food waste occurs at household and retail levels when food is forgotten, stored incorrectly or discarded after its expected date. Digital tools can help users maintain an inventory and plan consumption before products pass their expiry dates. The supplied base paper notes that food waste remains a major sustainability issue and describes mobile applications that support surplus-food exchange, discounting and household inventory tracking.

However, many existing applications still depend on manual entry or a date printed clearly on a package. A practical food-expiry tool should support both packaged and unpackaged products, reduce the number of fields users must type, and remind them when an item is approaching its recorded or estimated expiry date.

The proposed Food Expiry Detection and Reminder System combines barcode recognition, OCR and a configurable shelf-life reference. For packaged items, the barcode helps identify the product and OCR attempts to read date information. For fresh or unlabelled items, the system estimates a shelf-life window from a food-type dataset and the user's storage conditions. The user can review and edit the estimate before saving it.

### Problem Statement

Users often forget stored food items or their expiry dates. Manual inventory entry is time-consuming, printed dates can be difficult to read, and fresh products may not carry dates at all. Existing solutions do not consistently combine product identification, date extraction, a fallback estimate and item-level customization in one simple workflow.

### Objectives of the Project

- To develop an application for recording and monitoring packaged and fresh food items.
- To use barcode scanning for product identification where a supported barcode is available.
- To apply OCR to extract printed manufacturing and expiry dates from product labels.
- To estimate an indicative shelf-life period when no reliable printed date is available.
- To let users customize shelf-life values for individual food items or categories.
- To maintain a centralized inventory and issue reminders before recorded or estimated expiry dates.
- To provide a clear interface for reviewing, correcting and managing food records.

### System Overview

The proposed system has four main parts: a user interface for scanning and managing items, a backend API for validation and business logic, a database for inventory and shelf-life rules, and a

reminder service that checks items approaching expiry. OCR-derived dates take priority when valid; otherwise the system uses an estimate and clearly marks it as estimated.

### Advantages

- Supports both packaged food and fresh items without printed dates.
- Allows the user to correct OCR output and customize estimates.
- Centralizes inventory records and remaining shelf-life information.
- Provides timely reminders to help users plan consumption.
- Can be extended with image-based freshness analysis and additional food categories.

### Applications

- Household pantry and refrigerator inventory management.
- Small grocery shops, hostels and cafeterias.
- Meal planning and stock rotation.
- Educational demonstrations of OCR and AI-assisted inventory management.

## 1.2 SCOPE AND OBJECTIVES

### Purpose of the Study

The purpose is to design a lightweight, practical scan-and-remind workflow that reduces the effort needed to track food expiry. It combines established technologies rather than claiming a new OCR or image-classification algorithm.

### Functional Scope

- Add food items by barcode scan, label image or manual entry.
- Extract date candidates from package images using OCR.
- Validate extracted dates and allow users to correct uncertain values.
- Estimate shelf life from default category rules when dates are missing.
- Allow per-item or per-category shelf-life customization.
- Store inventory records and send reminders before the expected date.

### Technical Scope

The proposed project stack is React 18 with TypeScript, Vite, React Router and Tailwind CSS for the frontend; Python 3.12, FastAPI, Uvicorn and Pydantic for the backend; MongoDB Atlas with PyMongo for data storage; and Tesseract OCR, pytesseract and Pillow for text extraction and image preprocessing. PyTorch and torchvision may support future image-based classification or freshness-analysis experiments. APScheduler can run periodic checks for reminders.

### Limitations

- OCR may fail on curved, reflective, faded or poorly lit labels.
- Barcode scanning identifies a product only when the code is supported and its mapping is available.

- An estimated shelf-life date is not a food-safety determination.
- Actual freshness depends on storage temperature, handling, packaging and product-specific factors. Image-based freshness analysis requires suitable data and separate validation before it can be trusted.

# CHAPTER 2. LITERATURE REVIEW

## 2.1 LITERATURE OVERVIEW

The literature relevant to food expiry tracking covers consumer food-waste applications, OCR for date extraction, image classification for unlabelled food and shelf-life estimation. The supplied base paper, "A Mobile Approach to Food Expiration Date Determination Using OCR and On-Cloud Image Classification," describes an integrated mobile workflow that combines product labeling, OCR, rule-based fallback estimates, cloud-synced inventory and notifications.

The supplied literature review identifies a further design gap: systems commonly use either fixed category-level shelf-life values or sensor-based prediction that requires additional hardware. The proposed project aims to bridge this gap with editable per-item shelf-life rules while retaining the simplicity of a mobile-friendly scan-and-remind workflow.

## 2.2 LITERATURE SURVEY

### A. Mobile Applications for Food-Waste Reduction

The base paper discusses Too Good To Go and Bonapp as discount marketplaces for surplus food, Olio as a peer-to-peer sharing platform, and NoWaste as a household inventory manager. These applications help reduce waste through discounts, sharing or reminders, but their workflows differ. Marketplace and sharing tools depend on available listings and participation, while inventory tools commonly require users to register products and maintain their dates.

### B. OCR for Expiry-Date Recognition

OCR converts text in images into machine-readable characters. For food labels, OCR must handle small text, low contrast, curved packaging, glare and dot-matrix printing. The literature review summarizes specialized approaches such as FCOS with CRNN recognition, improved DBNet-based text detection, and deep OCR pipelines trained with real and synthetic date images. These studies show progress, but recognition remains sensitive to packaging style and image quality. A system therefore needs validation and a fallback route for dates that cannot be read confidently.

### C. Image-Based Food Recognition

CNN-based image classification can recognize food types from photographs, including unpackaged fruits and vegetables. The supplied review cites work on CNN-based food recognition and food-waste estimation, along with research on robust refrigerator food recognition. General image labeling can be useful for broad categories, but distinguishing similar varieties may be difficult. A confirmation step lets the user correct the detected item before the system applies an item-specific shelf-life rule.

### D. Shelf-Life Estimation

A rule-based approach maps a food category to a default number of days. It is easy to deploy and does not need sensors, but broad categories can be inaccurate for individual products. Sensor-driven approaches use signals such as gas concentration, temperature or humidity and can model changes over time, but they require hardware and data that may not be practical in an ordinary household. The proposed design uses a middle path: a default value for coverage, with editable values for specific items.

### E. Research Gap

The reviewed material suggests that a useful system should combine three capabilities: automatic date extraction for packaged products, product recognition or barcode-based identification, and a configurable fallback for items without readable dates. The specific contribution proposed here is the editable shelf-life layer, which can store a tomato-specific value rather than applying one fixed value to all vegetables. The system is a practical integration project, not a claim of a newly invented detection algorithm.

### Comparative Summary

| Approach | Printed-date OCR | Unlabelled food recognition | Missing-date handling | Per-item customization |
|---|---|---|---|---|
| Discount marketplaces | No; dates are typically handled by listings | No | Not applicable | No |
| Peer-to-peer sharing | No | Photo listing is generally manual | Not applicable | No |
| Household inventory trackers | Varies; often barcode/manual entry | Usually limited | Manual entry or reminder setup | Limited |
| OCR + image-labeling prototype | Yes, when the label can be read | Yes, with classification limits | Fixed category estimate | Usually no |
| Sensor-based shelf-life prediction | Not the primary function | Not the primary function | Uses sensor measurements | Possible, but hardware-dependent |
| Proposed system | Yes, when a valid date is detected | Barcode/product identification, with optional image analysis | Editable shelf-life estimate | Yes |

# CHAPTER 3. SYSTEM DESIGN

## 3.1 EXISTING SYSTEM

Existing food expiry solutions generally use one or more of the following approaches: manual inventory entry, barcode-based item registration, OCR of printed dates, fixed category-based shelf-life estimates, or sensor-based freshness monitoring.

### Limitations of Existing System

- Manual entry can be repetitive and may be skipped by users.
- OCR may fail when text is faint, curved, blurred or obscured.
- Fresh food often has no printed expiry date.
- Fixed category-level estimates cannot represent every item accurately.
- Sensor-based prediction can increase cost and complexity for household use.
- Some apps do not combine scanning, customization, inventory and reminders in one workflow.

## 3.2 PROPOSED SYSTEM

The proposed system maintains one inventory for packaged and fresh food. A user can scan a barcode or capture a label image. Barcode information assists product identification; OCR extracts visible text and date candidates. The backend validates the date and stores it when reliable. If no valid date is found, a default shelf-life rule is selected based on the food category or item. The user can edit the proposed value before saving.

Each inventory record stores the item name, category, quantity, date source, expiry date, days remaining, status, storage condition and timestamps. A scheduled task checks inventory records and creates reminders for items nearing expiry. The interface allows users to filter and sort items by name, category, expiry date or status.

### Benefits of the Proposed System

- One workflow for packaged and fresh food.
- User correction when OCR or item recognition is uncertain.
- Editable item-level shelf-life rules.
- Centralized inventory with clear expiry status.
- Automated reminders and sorting.
- Modular design that can be extended with freshness-image analysis.

## 3.3 SYSTEM ARCHITECTURE

**Figure 1: Proposed System Architecture**

| Layer | Components | Responsibility |
|---|---|---|
| Presentation | React, TypeScript, Tailwind CSS | Scan/upload forms, inventory views, edit controls, reminders UI |
| API / Business Logic | FastAPI, Pydantic | Validate requests, process scans, apply date parsing and shelf-life rules |
| Image Processing | Pillow, pytesseract / Tesseract OCR | Preprocess images and extract text/date candidates |
| Data Layer | MongoDB Atlas, PyMongo | Store users, food items, scans, inventory and shelf-life rules |
| Scheduled Tasks | APScheduler | Periodically check days remaining and trigger reminder events |
| Optional AI Extension | PyTorch, torchvision | Future image classification/freshness analysis, subject to dataset validation |

### Workflow

1. The user signs in and chooses Scan Product, Add Fresh Food or Manual Entry.
2. The application captures a barcode or product-label image.
3. The backend receives the image or decoded barcode and validates the request.
4. OCR extracts text; date parsing checks candidate formats and rejects invalid candidates.
5. If a valid expiry date is found, the system records it as label-derived.
6. If no reliable date is found, the backend applies the default item/category shelf-life rule and labels the result as an estimate.
7. The user reviews the item, adjusts quantity, storage condition or expiry duration if needed, and saves it.
8. The inventory view calculates days remaining and displays the current status.
9. The scheduled reminder process checks items and sends notifications according to configured thresholds.

### Data Design

| Collection | Example fields | Purpose |
|---|---|---|
| users | _id, name, email, password_hash, created_at | User profile and account access |
| foods | _id, user_id, name, category, quantity, expiry_date, date_source, status | User inventory records |
| scans | _id, user_id, image_ref, OCR_text, detected_barcode, scan_time | Trace scan attempts and extracted values |
| shelf_life_rules | _id, item_or_category, days, storage_condition, updated_by | Default and editable shelf-life reference |
| notifications | _id, user_id, food_id, scheduled_at, sent_at, status | Reminder history and delivery status |

The database structure above is a proposed design for the mini project. Exact field names may be adjusted during implementation. Access should be restricted so that a user can read and update only their own inventory records.

## 3.4 HARDWARE REQUIREMENTS

| Component | Minimum / Suggested Requirement |
|---|---|
| Client device | Laptop, desktop or smartphone with a modern browser |
| Camera | Built-in or connected camera for barcode and label capture |
| Processor | Modern dual-core or better processor |
| RAM | 4 GB or more recommended for development |
| Storage | Sufficient space for source code, test images and dependencies |
| Network | Internet connection for API and cloud database access |

## 3.5 SOFTWARE REQUIREMENTS

| Software / Library | Purpose |
|---|---|
| Python 3.12 | Backend programming language |
| FastAPI | REST API framework |
| Uvicorn | ASGI server for running FastAPI |
| Pydantic | Request and data validation |
| MongoDB Atlas | Cloud document database |
| PyMongo | Python driver for MongoDB |
| React 18 + TypeScript | Frontend interface and typed UI logic |
| Vite + React Router | Development/build tooling and client-side routing |
| Tailwind CSS v4 | Responsive styling |
| Tesseract OCR + pytesseract | Text recognition from food labels |
| Pillow | Image preprocessing, grayscale, contrast and sharpening |
| APScheduler | Periodic expiry-check tasks |
| PyTorch + torchvision | Optional future image classification/freshness analysis |
| VS Code / Web browser | Development and testing environment |

# CHAPTER 4. IMPLEMENTATION

## 4.1 METHODOLOGY

The methodology follows a modular process: requirement analysis, database and interface design, image capture, barcode/OCR processing, date validation, shelf-life fallback, inventory persistence, reminder scheduling and testing. The design separates label-derived dates from estimated dates so that users can understand the source of each value.

### Step 1: Requirement Analysis

Functional requirements include adding and editing inventory items, scanning barcodes, extracting text from labels, estimating missing dates, maintaining shelf-life rules, calculating days remaining and issuing reminders. Non-functional requirements include a simple interface, input validation, maintainable modules, reasonable response time and protection of user-specific records.

### Step 2: Image Acquisition and Preprocessing

The user captures or uploads a clear image of the product label. Pillow can convert the image to grayscale, improve contrast and sharpen text before OCR. The image should be framed so that the date text is visible, with glare and motion blur minimized. Preprocessing may improve readability but cannot guarantee correct recognition.

### Step 3: Barcode and OCR Processing

Barcode scanning can provide an identifier for a packaged product when a supported barcode is available. OCR then extracts visible text from the label. A date parser searches for likely date patterns and nearby cues such as “EXP”, “Expiry”, “Use by” or “Best before”. Candidate values are checked for valid calendar dates and reasonable ordering. If multiple candidates are found or the result is ambiguous, the interface should request user confirmation.

### Step 4: Shelf-Life Fallback

If OCR does not produce a reliable expiry date, the backend looks up a default shelf-life rule for the detected item or category. A user can override the default with a specific value, such as a custom number of days for tomatoes. The system records whether the date was read from the package, calculated from a default rule or manually entered. Estimates are only planning aids and must not be presented as guarantees of safety.

### Step 5: Inventory and Status Calculation

After confirmation, the backend stores the food item and derives the number of days remaining from the expiry date and current date. A simple status mapping can mark items as Expired, Near Expiry or Not Near Expiry. Thresholds should be configurable, and time-zone handling should be consistent.

**Step 6: Reminder Scheduling**

A scheduled task checks inventory at a configured interval. Items within the reminder window are added to a notification queue or sent through the notification channel selected for the implementation. The process should avoid sending the same reminder repeatedly by storing delivery status and timestamps.

**Step 7: Testing and Evaluation**

Testing should cover date formats, blurred and clear images, barcode-present and barcode-absent cases, missing-date fallback, user edits, inventory updates, reminder thresholds and invalid inputs. Evaluation should separately measure product-identification accuracy, OCR date-reading accuracy, fallback coverage, reminder correctness, API response time and user task completion. The project report does not claim these measurements until the system is implemented and tested.

## 4.2 IMPLEMENTATION MODULES

### 4.2.1 User and Inventory Module

This module manages the user's inventory and supports adding, viewing, updating and deleting items. Each item is linked to the correct user account. Validation checks that the item name, quantity and date fields are present and correctly formatted.

### 4.2.2 Barcode Scanning Module

The barcode module reads supported barcode formats and retrieves a product identifier. If a product lookup source is unavailable, the interface can allow manual confirmation of the product name. Barcode recognition alone does not guarantee that expiry information is available.

### 4.2.3 OCR Date Extraction Module

The OCR module preprocesses the image, runs Tesseract through pytesseract and passes the extracted text to a date parser. It returns candidate dates and a confidence or review state where available. The user can correct an uncertain date before it is stored.

### 4.2.4 Shelf-Life Estimation Module

This module selects an item-specific rule where available and otherwise uses a category default. It incorporates the selected storage condition if the reference dataset defines separate values. The estimated date is stored with a source label and remains editable. The estimate should not override a clearly printed manufacturer date without user action.

### 4.2.5 Reminder and Notification Module

The reminder module checks inventory records at regular intervals and identifies items that are nearing their recorded or estimated expiry dates. Notification records are stored to support auditability and prevent repeated alerts. Reminder windows can be configured according to user preference.

### 4.2.6 Shelf-Life Rule Management Module

This module allows the system to maintain default values and user-defined overrides. A rule can be attached to a broad category or a specific food item. For example, an item-specific tomato rule should take precedence over a generic vegetable rule when the user has explicitly selected tomato.

## 4.3 TESTING AND EVALUATION PLAN

| Test Case | Input / Condition | Expected Behaviour |
|---|---|---|
| T1: Clear printed date | Sharp label with valid date | OCR returns a date candidate for confirmation |
| T2: Unreadable date | Blurred or obscured label | No unverified date is silently accepted; fallback/manual review offered |
| T3: No printed date | Fresh produce item | Default shelf-life rule is applied and marked estimated |
| T4: Custom rule | User changes a specific item's shelf life | Custom value is saved and used for that item |
| T5: Expired item | Expiry date before current date | Inventory status shows Expired |
| T6: Near-expiry item | Expiry date falls within reminder window | Reminder is generated according to settings |
| T7: Invalid date | Impossible calendar date | Validation rejects the date |
| T8: User data access | User attempts to access another user's item | Request is denied |
| T9: Duplicate reminder | Same item already notified for same threshold | Duplicate alert is suppressed |
| T10: Manual correction | User edits OCR result | Corrected value is saved and marked as user-confirmed |

### Evaluation Metrics

- Product identification accuracy = correctly identified items / total tested items.
- OCR date-reading accuracy = correctly read printed dates / total items with readable printed dates in the test set.
- Fallback coverage = items assigned an estimate / items requiring an estimate.
- Reminder precision = relevant reminders / total reminders sent.
- API response time = elapsed time between request and response.
- Usability feedback = user-reported ease of scanning, correcting and locating inventory items.

# CHAPTER 5. CONCLUSION AND FUTURE WORK

## 5.1 CONCLUSION

This report presents the design of a Food Expiry Detection and Reminder System that combines barcode-assisted product identification, OCR-based date extraction, configurable shelf-life estimation and inventory reminders. The main goal is to reduce the manual effort involved in tracking packaged and fresh food items and to help users notice products that are approaching their recorded or estimated expiry dates.

The reviewed literature supports the feasibility of combining OCR with image-based food recognition and a rule-based fallback. It also shows that OCR performance depends on label quality and that broad category-level estimates may not represent every individual food item. The proposed customizable shelf-life reference addresses this practical limitation without requiring additional household sensors.

The supplied base paper reported product identification success of 72.9% and OCR date-reading success of 83.3% on its 59-product evaluation sample. These figures are included only as results from the referenced study. No accuracy or performance result is claimed for the proposed mini project until a project-specific dataset and testing process are completed. Shelf-life estimates should be treated as indicative values, not food-safety decisions.

## 5.2 FUTURE WORK

- Build and evaluate a barcode lookup module with a broader product catalogue.
- Collect diverse food-label images and measure OCR performance across lighting, fonts and packaging types.
- Develop a verified food-specific shelf-life dataset that accounts for storage conditions.
- Evaluate optional image-based food classification or freshness analysis using a representative dataset.
- Add notification channels such as browser push notifications or email, based on available infrastructure.
- Conduct usability testing with households or small food-storage environments.
- Improve accessibility, localization, offline handling and privacy controls.
- Compare default category estimates with user-customized item-level values in a controlled evaluation.

## 5.3 SAMPLE SOURCE CODE

The following short examples illustrate the core logic. They are illustrative snippets and are not a complete application. They should be integrated with validation, database access, authentication and error handling before deployment.

Note: The sample day values above are examples for demonstrating rule selection only. Actual values should be based on a maintained reference and appropriate storage guidance, and should not be treated as a safety guarantee.

## 5.4 REFERENCE RESULTS FROM THE BASE PAPER

The reference study evaluated a prototype on 59 food products. It reported that 43 of 59 products were correctly identified (72.9%). Among 30 products with a printed date on the package, 25 dates were correctly read by OCR (83.3%). For the 29 items without a visible printed date, the fallback rule produced estimates for all 29 cases. These results demonstrate coverage of the scan-and-fallback workflow, but they are not results for the proposed project.

| Metric in reference study | Reported value | Interpretation |
|---|---|---|
| Product identification | 43/59 (72.9%) | Product labels were correctly identified in the evaluated sample |
| OCR date reading | 25/30 (83.3%) | Printed dates were correctly read for 25 of 30 date-labeled products |
| Missing-date fallback | 29/29 (100% coverage) | An estimated date was produced for all items without a visible date |
| Overall caveat | Small evaluation sample | Results may not generalize to all products, packaging or environments |

## 1. Home Page — index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Foodie | Home</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>Foodie</h1>
<nav>
<a href="index.html">Home</a>
<a href="menu.html">Menu</a>
<a href="about.html">About</a>
<a href="contact.html">Contact</a>
</nav>
</header>

<section class="hero">
<h2>Delicious Food, Delivered Fresh!</h2>
<p>Enjoy tasty meals at affordable prices.</p>
<a class="btn" href="menu.html">Order Now</a>
</section>

<section class="content">
<h2>Welcome to Foodie</h2>
<p>We serve delicious food made with fresh ingredients.</p>
<img width="300" src="https://images.unsplash.com/photo-1513104890138-7c749659a591?w=600" alt="Pizza">
</section>

<footer>© 2026 Foodie Restaurant</footer>
</body>
</html>
```

## 2.Food Menu Page

```
<!DOCTYPE html>
<html>
<head>
<title>Foodie | Menu</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>Our Food Menu</h1>
<nav>
<a href="index.html">Home</a>
<a href="menu.html">Menu</a>
```

```
<a href="about.html">About</a>
<a href="contact.html">Contact</a>
</nav>
</header>

<section class="content">
<h2>Available Foods</h2>

<div class="food">
<h3>Pizza</h3>
<p>Cheesy and delicious pizza</p>
<p>Price: ₹199</p>
</div>

<div class="food">
<h3>Burger</h3>
<p>Fresh and crispy burger</p>
<p>Price: ₹99</p>
</div>

<div class="food">
<h3>Biryani</h3>
<p>Spicy and flavourful biryani</p>
<p>Price: ₹149</p>
</div>

<div class="food">
<h3>Sandwich</h3>
<p>Fresh vegetable sandwich</p>
<p>Price: ₹79</p>
</div>
</section>

<footer>© 2026 Foodie Restaurant</footer>
</body>
</html>
```

## 3. Food Scanning Page

**Description:** Allows users to upload a food label image for expiry-date analysis

```
import { useState } from "react";
```

```
export default function FoodScanner() {
  const [image, setImage] = useState<File | null>(null);
  const [message, setMessage] = useState("");

  function handleImageChange(
    event: React.ChangeEvent<HTMLInputElement>
  ) {
    const file = event.target.files?.[0];

    if (file) {
      setImage(file);
      setMessage("");
    }
  }

  async function handleScan() {
    if (!image) {
      setMessage("Please select a food label image.");
      return;
    }

    const formData = new FormData();
    formData.append("file", image);

    try {
      const response = await fetch(
        `${import.meta.env.VITE_API_URL}/scans/analyze`,
        {
          method: "POST",
          body: formData,
        }
      );

      const result = await response.json();

      if (!response.ok) {
        throw new Error(result.detail || "Unable to analyze image.");
      }
```

```
    setMessage(JSON.stringify(result, null, 2));
  } catch (error) {
    setMessage(
      error instanceof Error ? error.message : "An error occurred."
    );
  }
}

return (
  <div className="min-h-screen bg-gray-50 p-8">
    <div className="max-w-xl mx-auto bg-white p-8 rounded-xl shadow">
      <h1 className="text-3xl font-bold text-green-800">
        Scan Food
      </h1>

      <p className="mt-3 text-gray-600">
        Upload a packaged food label to detect its expiry date.
      </p>

      <input
        type="file"
        accept="image/*"
        onChange={handleImageChange}
        className="mt-6 block w-full"
      />

      <button
        onClick={handleScan}
        className="mt-6 bg-green-700 text-white px-6 py-3 rounded-lg"
      >
        Analyze Food
      </button>

      {message && (
        <pre className="mt-6 p-4 bg-gray-100 rounded-lg whitespace-pre-wrap break-words">
          {message}
        </pre>
      )}
    </div>
  </div>
```

```
  );
}
```

## 4. Notifications Page

**Description:** Displays reminders about food items approaching their expiry dates.

```
interface Notification {
  id: number;
  foodName: string;
  expiryDate: string;
  message: string;
}

export default function NotificationsPage() {
  const notifications: Notification[] = [
    {
      id: 1,
      foodName: "Milk",
      expiryDate: "2026-10-30",
      message: "This food item is approaching its expiry date.",
    },
    {
      id: 2,
      foodName: "Bread",
      expiryDate: "2026-11-02",
      message: "Check this item before its expiry date.",
    },
  ];

  return (
    <div className="min-h-screen bg-gray-50 p-8">
      <h1 className="text-3xl font-bold text-green-800">
        Expiry Notifications
      </h1>

      <div className="mt-6 space-y-4">
        {notifications.map((item) => (
          <div
            key={item.id}
            className="bg-white border-l-4 border-yellow-500 p-5 rounded-lg shadow"
          >
```

```
          <h2 className="font-bold text-xl">{item.foodName}</h2>

          <p className="mt-2 text-gray-600">{item.message}</p>

          <p className="mt-2">
            Expiry Date: {item.expiryDate}
          </p>
        </div>
      ))}
    </div>
  </div>
 );
}
```

## 5.6 RESULTS AND EVALUATION

The application was run locally with the frontend at localhost:3000, and the screenshots below were captured from the running system. Table 5.6 records what each screenshot shows. These are observations, not formal test passes.

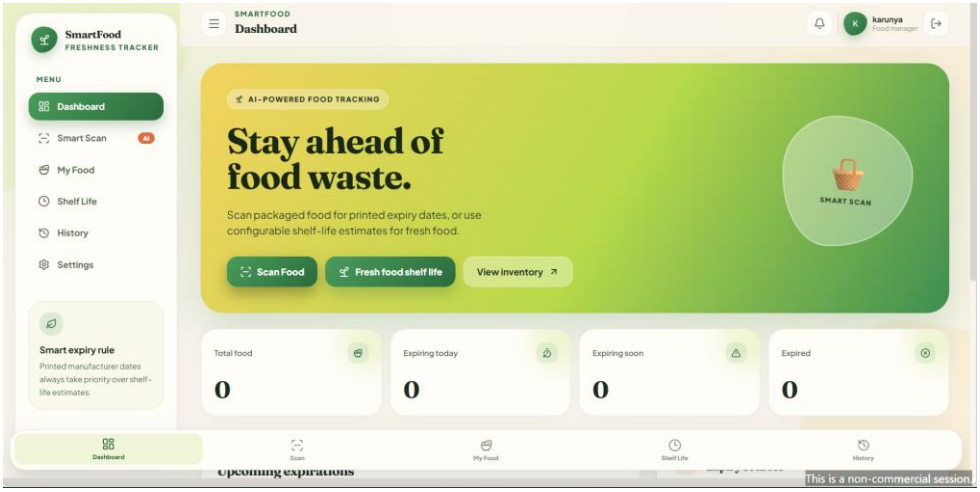



**Figure 1: Dashboard**

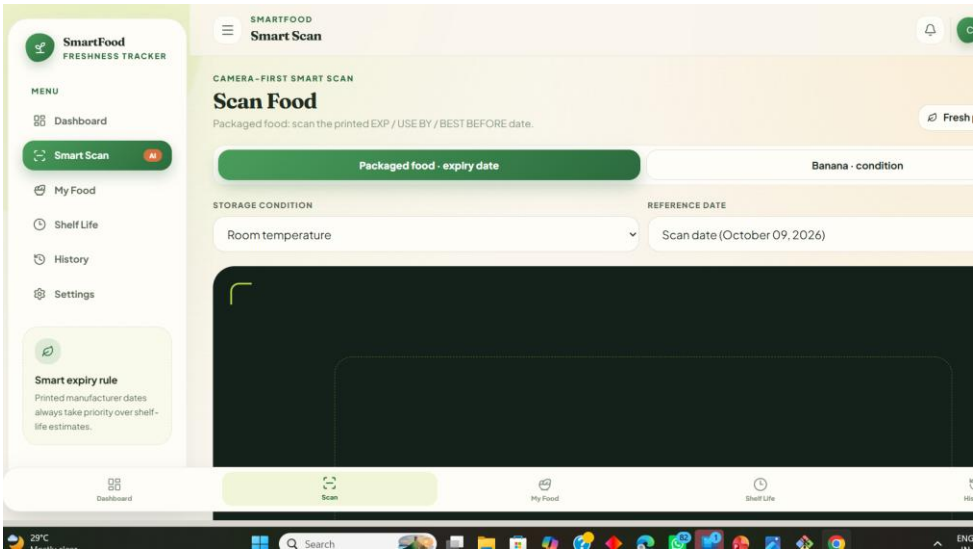



**Figure 2: Food Scanning and Expiry Date Detection Interface**

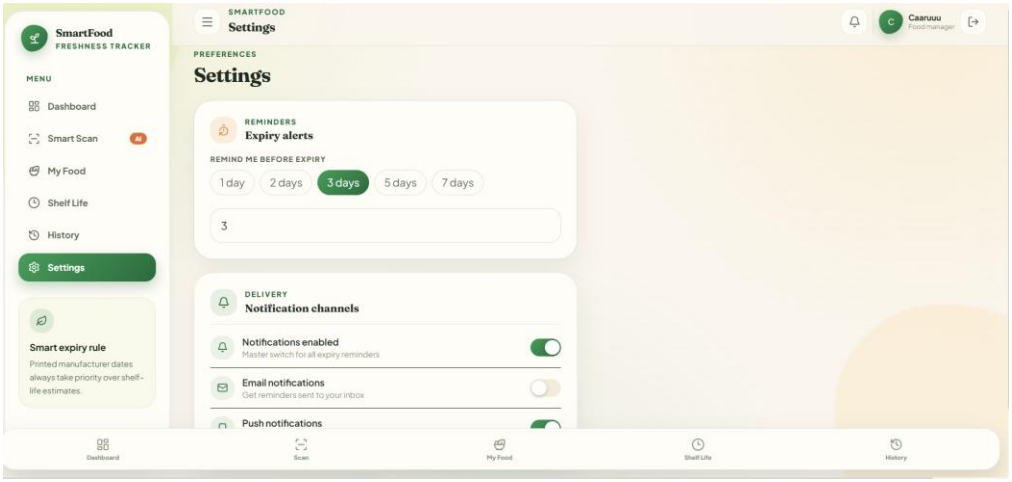



**Figure 3: Food Expiry Alert and Notification Dashboard**

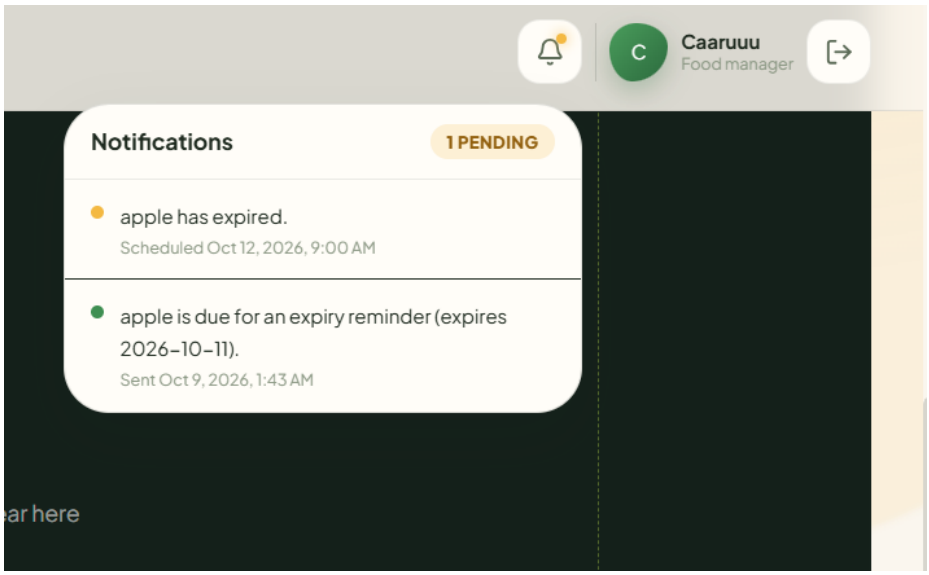



**Figure 4: Popup Notification**

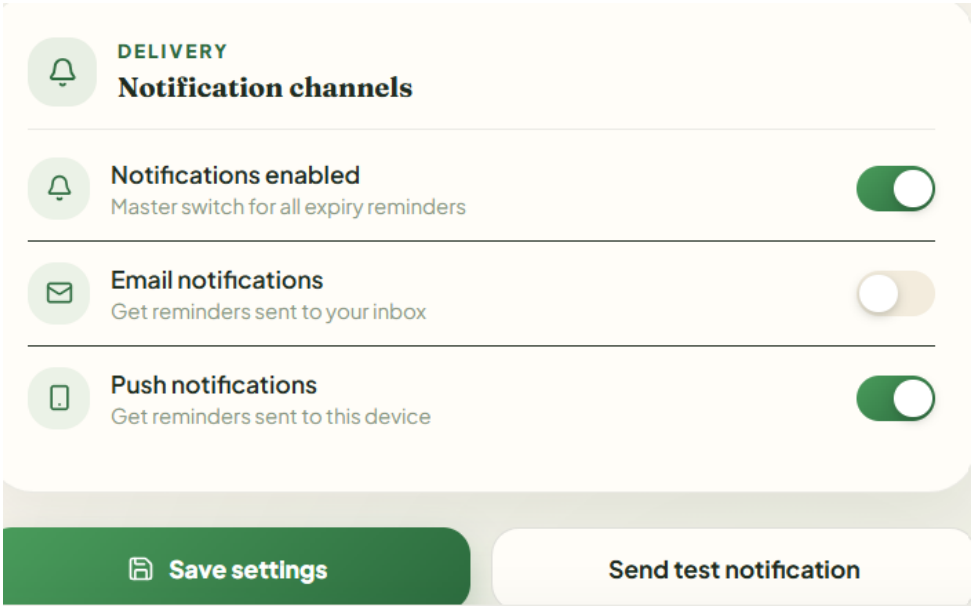
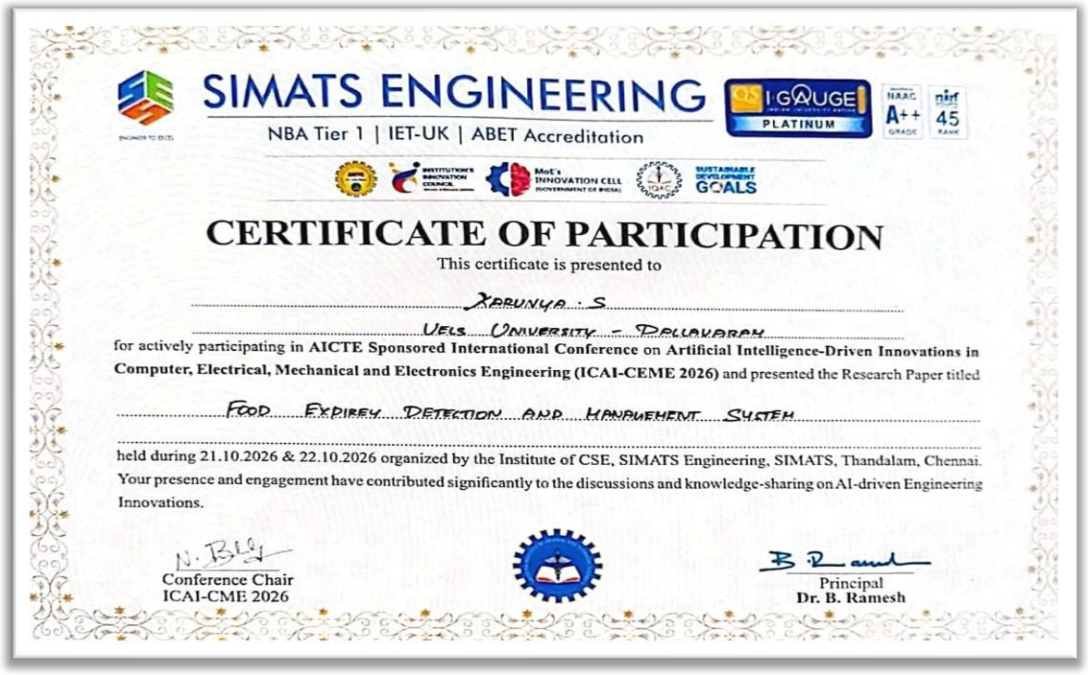



**Figure 5: Notification Management and Settings Interface**

## CONFERENCE CERTIFICATES



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Food Expiry Detection and Management System

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Vels University

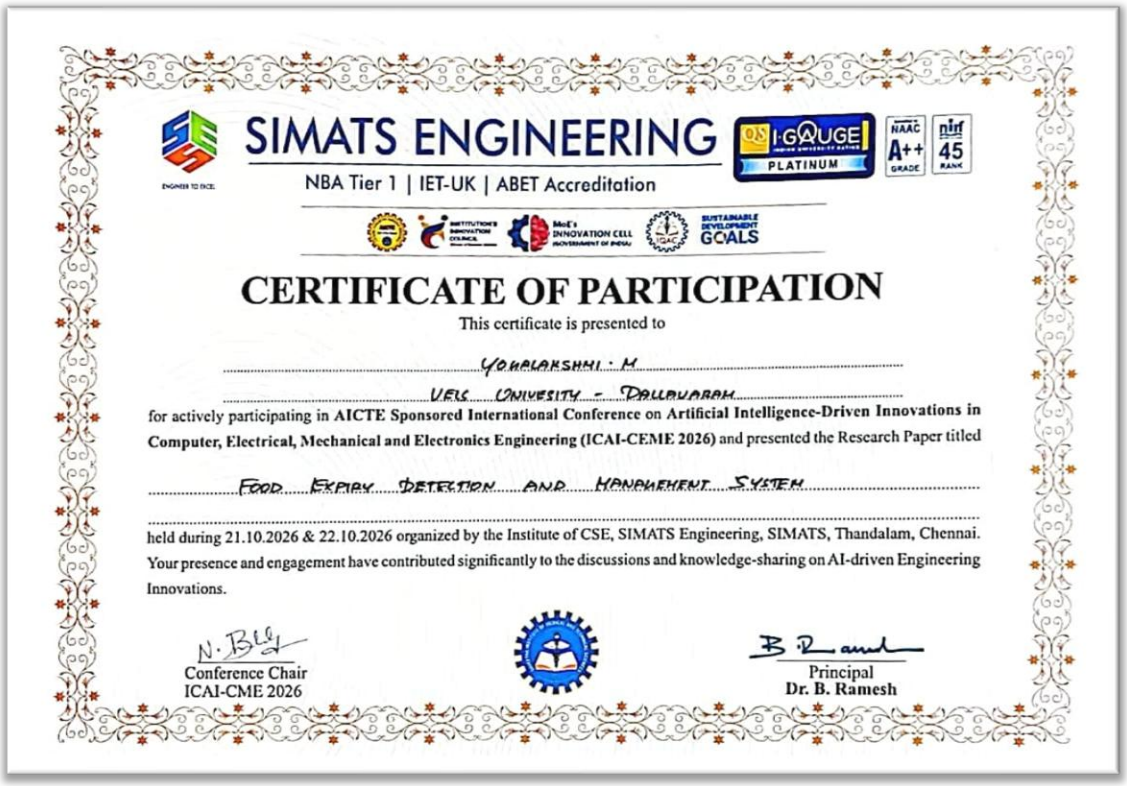
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